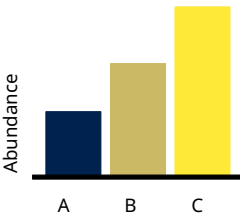
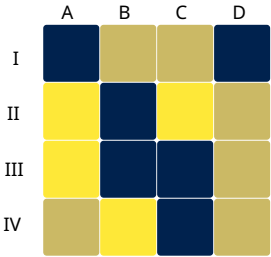


A) Theoretical background

Traditional metabarcoding
(Abundance / presence-absence matrices)

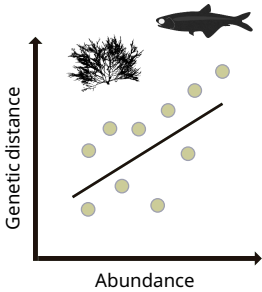


Hypothesis:
Abundance $\neq$ Genetic distances
(reads) (nucleotide diversity)



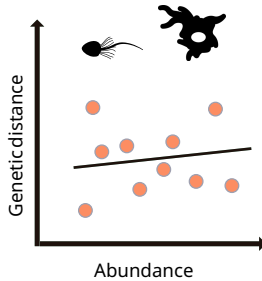
Multicellular phyla

- ↑ Longer life cycles
- ↑ Sexual reproduction
- || Higher correlation



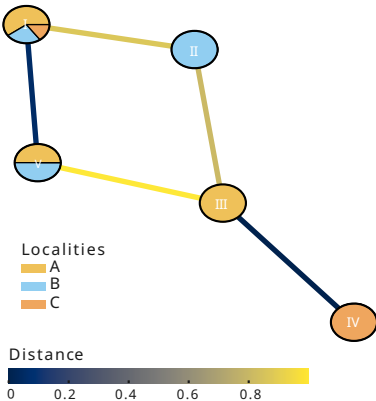
Unicellular phyla

- ↓ Shorter life cycles
- ↓ Sexual reproduction
- || Lower correlation



B) Methodological implementation for population genetics in metabarcoding studies

1. Haplotypic network



2. Connection table

Haplotype	Haplotype	Locality	Locality	Genetic distance	Geographic distance
I	I	A	B	0	0.9
I	I	A	C	0	0.2
I	II	A	B	0.6	0.9
I	II	B	C	0.6	0.6
I	II	B	B	0.6	0
I	V	A	A	0.2	0
I	V	A	B	0.2	0.9
I	V	A	C	0.2	0.2
I	V	B	C	0.2	0.6
I	V	B	B	0.2	0
⋮	⋮	⋮	⋮	⋮	⋮

3. Localities network

